

RENANCE

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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 121 — Past Questions & Materials  
**by Renance**

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Course: MTH 121  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 1  
Levels: 100

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# Paper 1 of 1 — MTH 121 PAST QUESTION 2015 AND 2016

Level 100 • Second Semester • PastQuestion

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“ghee. FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.

x COLLEGE OF SCIENCE

ROE. DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

SECOND SEMESTER EXAMINATION 2015/2016 SESSION

COURSE TITLE & CODE; GENERAL MATHEMATICS II (MTH 121) = TIME: 2 thes

TYPE D

nstructio ms: Anwser all questions on the Answer booklet by writing the preferred option eg, Lor B, Do all your rough work from the back pages of the answer booklet. Submit both your juestion paper and answer booklet at the end of the examination.

Yame:

Mtric No. Dept.

1. What TYPE is your question paper? A. TYPE A B. TYPE B C. TYPE CD. TYPED

5)  $dx$  A.  $\ln@e=-3)+\phi$  B.  $2\ln@? -6x+2)4\phi$  C.  $2n@=3)+eD$ .  $\tan@? --6x+$

$2D+e$

S.Integmts JeQt-aypde A Set-1F+eB. EG -+e C SO%-t4 te D.

eeoky

4. Solve  $f = - - \sim 5$  A  $\ln Qx+1)+cB$ .  $\sin aes +x-2)4cD$ .  $2-A4\phi$

$_{\sin 2x} \_ \cos 2x \_ \cos 2x \sin 3x$

eye fianae foe on = a teBi= --- oO oD

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$Z 3x^2$  we Ge ae ee (ean 2

6.  $fx2e dx$  A.  $EiSye$  B.  $-5xe +eC$ , ram =e Ai D.  $7 (E+2)+$

7. Find the integral rz

8. Integrate  $[ e^{?}*8dx$  A.

Fi  $\phi B$ .  $3e^3 4\phi C$

city

is ae si

9. Solve  $f \sin? x \cos xax$  A.  $-\cos? xsinx +2B$ .

$-\sin tx$  Cte is

el sin casi 16) &

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ne tae Eth+cB. 205 +06, Sete, ite

11. Evaluate  $f - z \text{ mat } \sin(2) + e$ .  $\cosh^{\text{TM}} 'Bec \ln(x? + 25) D$ .  $z \tan^{\text{TM}} \text{ an}() + 0$

12. If  $y = x \sin x$ , find  $A$ .  $\sin x - \cos x$  B.  $\sin x + \cos x$  C.  $\sin x - x \cos x$  D.  $\sin x + x \cos x$

13. If  $y = (2x + 1)^2$ , find  $\frac{dy}{dx}$ . A.  $6(2x + 1)^2$  B.  $3(2x + 1)^2$  C.  $6(2x + 1)$  D.  $3(2x + 1)$

14. If  $y = 3 \cos 4x$ , equals  $A$ .  $6 \sin 8x$  B.  $-24 \sin 4x$  C.  $12 \sin 4x$  D.  $-12 \sin 4x$

15. If  $s = (2 + 3t)(5t - 4)$ , find  $\frac{ds}{dt}$  when  $t = 2$ . A. 0 units per sec B. 15 units per sec C. 22 units per sec D. 26 units per sec

16. Differentiate  $\sin x$ . A.  $x \cos x$  B.  $x \sin x$  C.  $-\cos x$  D.  $-\sin x$

a.  $\frac{dy}{dx}$

a.  $\frac{dy}{dx}$

17. Find the derivative of  $y = x^3 - 1$ .

x

A.  $x(x^3 - 1)$  B.  $3x(x^3 - 1)$  C.  $3x^2 - 1$  D.  $7x^9 - 5x^4$

18. Differentiate  $(x^3 - 4)$  with respect to  $x$ . A.  $4x^3 - 4$

$4x^3 - 4x - 4$

B

19. If  $y = (1 - 2x)^3$ , find the value of  $\frac{dy}{dx}$  at  $x = -1$ . A. -6 B. 5 C. -5 D. 27

20. Find the derivative of  $y = \sin(2x^3 + 3x - 4)$ . A.  $-\cos(2x^3 + 3x - 4)$  B.  $-(6x^2 + 3) \cos(2x^3 + 3x - 4)$

C.  $\cos(2x^3 + 3x - 4)$  D.  $(6x^2 + 3) \cos(2x^3 + 3x - 4)$

$2x^3 + 3x - 4$

21. Which of the following is true for the function  $f(x)$  defined by  $f(x) = 4x^2 + 1$  if  $x \geq 1$

and  $f(x) = 3$  if  $x < 1$ ?

A.  $f$  is continuous everywhere

B.  $f$  is continuous everywhere

C.  $f$  is continuous everywhere

D.  $f$  is continuous everywhere except at  $x = 1$

0

22. Find the values of  $k$  such that the function  $f(x) = \begin{cases} kx^2 - 1 & \text{if } x \geq 1 \\ 2x - 1 & \text{if } x < 1 \end{cases}$  is continuous at  $x = 1$ .

(a)  $k = -1$

A.  $k = -1$  only B.  $k = -2$  only C.  $k = -1$  and  $1$  D.  $k = -2$  and  $2$

A.  $k = -1$  only B.  $k = -2$  only C.  $k = -1$  and  $1$  D.  $k = -2$  and  $2$

5.  $y$

$\lim_{x \rightarrow 2} (x^2 - 4)$

23. Suppose  $f(x) = \begin{cases} x^2 + m & \text{if } x \geq 2 \\ 2x - 1 & \text{if } x < 2 \end{cases}$ . Find a value for  $m$  such that the function is continuous at  $x = 2$ .

A.  $m = 2$

B.  $m = 8$  C.  $m = 1$  D.  $m = 3$

ees ; aa

= 1 C. 2 D. Not exist

: xo AI BS CO DE

) of producing a quantity  $x$  of a product is  $4x^2$

fl 3¢ Gost per unit quantity  $A(x)$  cyuals the total cost divided by the quantity produced  $x$ . Pind limy.4e.  $A(a)$ . A.5 B. 6 C. 5000 D, 5006

Sear i Bes)

27. Le  $f(a) = \{ ? \}$  Of BSS re cimaceeiaesy) A, G BY Ae a pe

eee Nena ee 4 eS

28.1f limy  $gx? + 2x + 1 = 16$ . Find  $d > 0$ . Av  $o = \min\{1, 5\}$  B.  $6 = \min\{1, 5\}$  CG 6s

$\min\{1, 5\}$  DD. 6  $\min\{1, 5\}$

29. Suppose  $\epsilon =$  and the limit of  $1 + x?$  is 5 a8  $x + -2$ . Determine the minimum vauue of  $6 > 0$  es yee ae ci

A.  $d6 = 1$  B.  $\epsilon =$  Th Cee, D. Not exist 5 aa) ak t Eat

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¢ Li, 49 —— A. termmate B. 4 €. 6 D. 8

31. 1f  $g@) = vx - 2--3$  what is the Dom( $g(x)$ ) ia) Rang( $g(x)$ )? A. (0,09) By [2, 09) C, (—1, 03)

D. [-3, 92)

32: Find the common domain for  $f(x) = 3$  and  $@(x) = se + 727$  A.0.8 B: 1.6 6.5 D8

33. Determine the domain and range of the following function  $y = y--2x + 3$  A.  $x53, y \$0$  B.

3 3

R27 V21CKS Vy SOD. ne Oy e3

34. Find the maximum value of  $y = -x^* + 4$  for all values of  $x$ ? A.16.2 C.0D.4 ah

35. Suppose  $f(x) = aa$  a what is the domain of  $f(x)$ ? :

a

A, (—0, -1) U (2,00) B. (—00, —1] U [2, 0) C. (—00,—4) U [2, 0) D. (—e, —1] U (2, co)

36, Suppose  $f(t) = 2 + 5¢ + 1$ , find the limit ast  $+ -3$  A.15 B.42 C.-41 D.-27

ade :

> A2B4C6D.8 ;

a A.0 B.1 C.-1 D. indeterminate

al

39. Find the limit as  $x$  tends to 0 from the left  $\lim, 9- ---$

37. Find the limit  $\lim, .2$

38. Compute  $\lim, \_, 3$

si

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ope bbe

40. Eval Beate eA Oe)

Ay . 7 7 tat 12 98

41. Find the area under the curve  $y = \frac{1}{(2x - 1)}$  on the interval  $1 \leq x \leq 8$ . A.  $\frac{1}{2} \ln 8$  B.  $\frac{1}{2} \ln 16$  C.  $\ln 8$  D.  $\ln 16$

. "x . 6 63 ot

42. Find the area under the curve  $y = e^x + e^{-x}$  on the interval  $0 < x < \ln 8$ . A.  $3B$ .  $e$

8

43. For  $n$  = number of people infected in hundreds and  $t$  = number of days, the rate of spread of a flu

infection is found to be  $S = 5 + 3y$ . How many people would we expect to become infected on day 273

a days 9 to 162 A. 10,900 B. 1,090 C. 90,100 D. 700

2

tes

44. Find a value  $c$  that satisfies the integral mean value theorem for  $f(x) = x^2$  on  $[-1, 1]$ . A.  $\frac{1}{2}$  B.  $\frac{1}{3}$  C.  $\frac{1}{4}$  D.  $\frac{1}{5}$

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1 v2

fetta 5 0 7 ie =

5. Evaluate the definite integral  $\int_0^1 x^2 dx$ . A.  $\frac{1}{3}$  B.  $\frac{1}{2}$  C.  $\frac{1}{4}$  D.  $\frac{1}{5}$

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x

Find the area of the region bounded by  $y = xe^{-x}$ ,  $y = x + 1$ ,  $x = 2$  and the  $y$ -axis. A.  $3.5$  B.

1.0 C. 3.0 D. 2.0

48. Determine the area of the region bounded by  $y = 2x^2 + 10$  and  $y = 4x + 16$ . A. 2 B.

3 ' 5 -

de 1 % a

49, Evaluate  $\int_0^1 x^2 dx$ . A.  $\frac{1}{3}$  B.  $\frac{1}{2}$  C.  $\frac{1}{4}$  D.  $\frac{1}{5}$

50, Evaluate the definite integral  $\int_0^1 x^2 dx$ . A.  $\frac{1}{3}$  B.  $\frac{1}{2}$  C.  $\frac{1}{4}$  D.  $\frac{1}{5}$

at  $\int_0^1 x^2 dx$ . A.  $\frac{1}{3}$  B.  $\frac{1}{2}$  C.  $\frac{1}{4}$  D.  $\frac{1}{5}$

oH caren 3 34 73  $\sim 3 \times 5 =$

52. Find the integral of  $f(x) = x^2$ . A.  $\frac{1}{3}x^3$  B.  $\frac{1}{2}x^2$  C.  $\frac{1}{4}x^4$  D.  $\frac{1}{5}x^5$

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dx

Salon

54. Solve f

56. Evaluate  $\int \sin x \cos x dx$  A.  $\frac{1}{2}$ :

57. Find the integral of  $f(z) dz$  A.

ae

38.

dx

Kets

55. integrate  $\int \cos(3x) dx$  A.  $\frac{1}{3} \sin 3x + C$  B.  $-\frac{1}{3} \sin 3x + C$

dx

Linge + ayreoR.  $\ln(e+2) + e$ ,  $\sin k$  (:

$x \cos x + e$  B.-

RoR. fins eC, lope Di

A, LinGe? =  $4x + 1$  be B. Ine

$\sim 2$ ) C.  $2 \ln 2$  4D.  $\sin^2 4x + 7$  +e

59. Integrate  $(2x-3)' de$  A.  $(x-3) + C$ , BOM og, BM 6p Ba te

60.  $(3x^3)$

6

dx A.5B.6C.7D.8 Kc

Ol.  $y = 8 \cos \theta$ , find  $\theta$  when  $x = 9.201$  ag

62, Find the derivative of  $(2 +$

63. What is the rate of change of the

A. Bir 1. 1 Gur

Of. Find the deri

$G5$ ,  $W_y = B \sin$

dy

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ify =

69. Find the rate of  $\odot$

B.  $4a^2$  C,  $24a^2$

70. Differentiate

Sand  $y = 3$  when  $\theta = 0$ , find  $y$  in te

$3 \text{ Dex?} - 3x^4$

the derivative of  $y$

$2 \sin 5x \cos 5x$  D. J

YO  $x$ ) with respect to  $x$ . A,  $6B = 3$  C.  $b = 6x$  1D.  $6x = |$   
 volume  $v$  of a hemisphere with respect to is radius  $r$  when  $r = 2$ ?  
 C. 20 1). dar

ye of the function  $y = 2x^4$   $Cx = 1$ ) at the point  $x$  LACIE BLL  $\phi$  FI

A

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Joust: dada) 12 sing

tox. A.  $10 \sin 5x$   $5x$  B.  $5 \sin$

ange Of the volume  $V$  of a sphere  $w$   $usr$  when  $r = ]$  A, 129

Gir

aaa Bye  $> 4$ ) with respect to  $x$ , A.  $4 \tan(x - 4)$  B.  $4(2x + 5)(4x - 3)$  C.  $(2x + 5)(2x - 13)$  1.  $lx$  &  $Syox$  1p

$ae = fe$  oo  $y$

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$a' < Gs$

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